A3-PS003-TC019

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Central traffic prediction is costly and puts everyone's travel data in one place

Centralised approach

- Every camera and sensor sends raw data to one central server
- Costs a lot of internet bandwidth
- All travel data in one place: a privacy risk

Federated learning

- Each junction trains a small model on its own data
- Shares only model updates, not raw data
- Raw travel data stays at the junction

We are building a simulator to test how well this works and how much data it sends

Background

Bengaluru's traffic is the motivation: TomTom Traffic Index 2025 [1]

74%	168	36 min
congestion level	hours lost per year in traffic	to drive just 10 km

- We live in Bengaluru and deal with this traffic every day
- Pune, Mumbai and Delhi face the same problem, so an approach that works across cities is useful

Literature & RFC Review

FedAvg (McMahan et al., 2017) is the main paper behind our work [2]

10–100×

fewer communication rounds reported for FedAvg

- Devices train locally and a server averages their models
- Cuts the number of communication rounds by 10 to 100 times
- Still works when data is very different across devices, as with traffic where every junction has its own pattern

Our focus: apply FedAvg to traffic data and measure accuracy against communication cost

Proposed Solution

Simulate federated learning on virtual junctions and compare with a central model

Data	IISc AIM BDE dataset split across 3 to 5 virtual junctions
Local models	Each junction trains a small prediction model on its own data
Aggregation	A central server combines the models using FedAvg
Time-based split	Train on earlier data, test on later data: the model never sees the future
Two scenarios	Similar data at every junction vs very different data at each junction
Baseline	One central model trained on all the data
Drivers	What drives traffic at each place: time of day, day of week, weather

Optimization Proposed

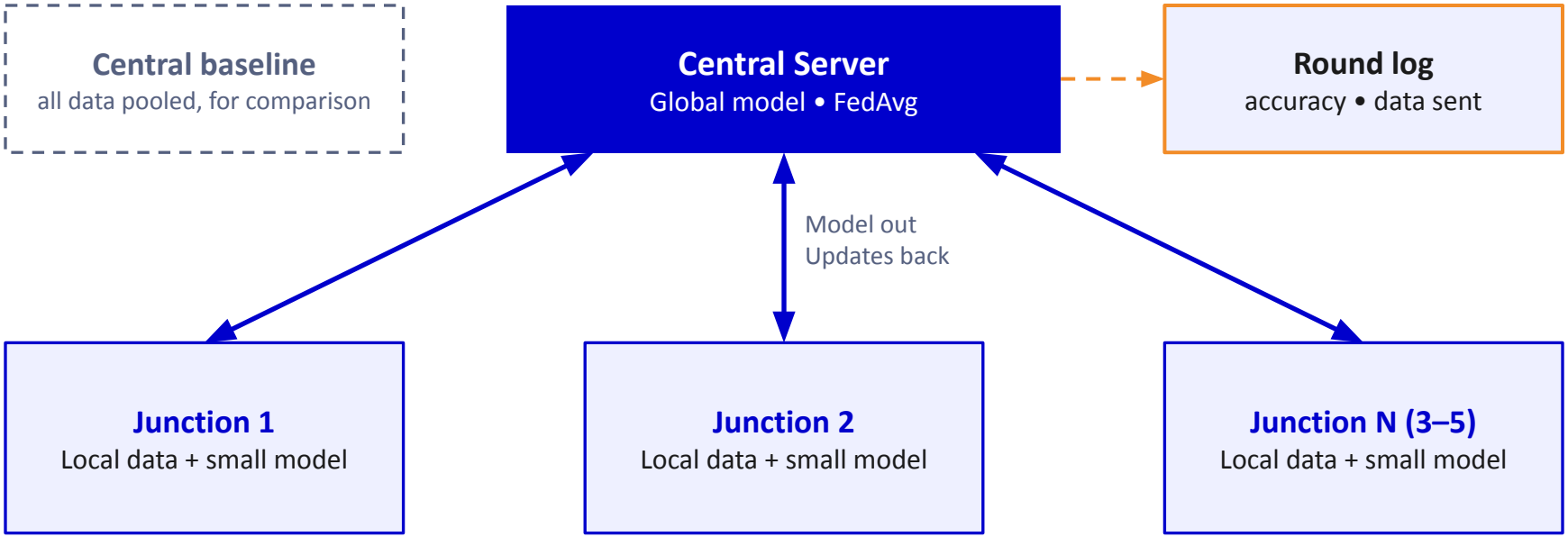
Central approach vs federated approach

Aspect	Central approach	Federated approach
Raw data	Sent from every junction to one server	Stays at each junction
What is sent	All the raw sensor and camera data	Only small model updates
Travel data privacy	All travel data in one place	Never pooled in one place
Other cities	Raw data from every city pooled centrally	Each city or zone joins as its own client

Each round, the simulator records accuracy and the amount of data sent, so the trade-off can be measured

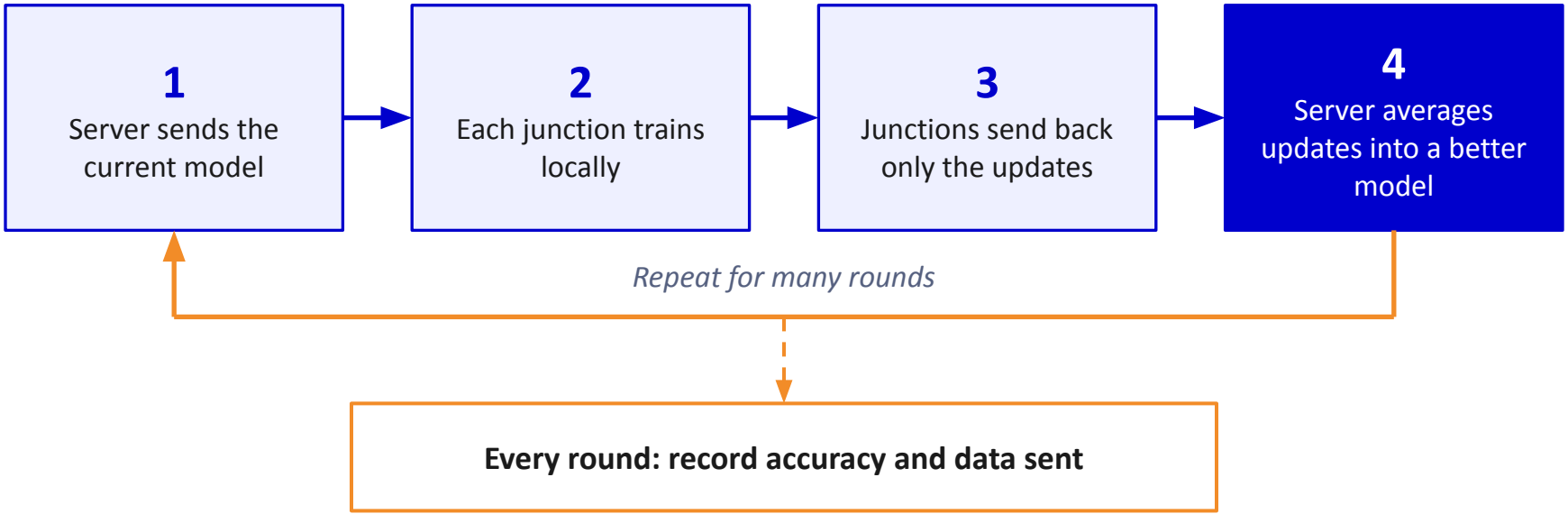
Solution Architecture & Design

Each junction keeps its data and model; only updates travel to the server



Data Flow Diagram

One federated round, repeated many times



Deployment Timeline

Milestones and who owns them

1 Set up

GitHub repo and first prototype

2 Data & baseline

Research, data preparation, time-based split, central baseline

3 Simulate

Prediction model, federated simulator, experiments

4 Report

Accuracy, data sent and learning speed; final report

Experiments run 3 times with different random seeds, so results are reliable

Bibliography

[1] TomTom. “TomTom Traffic Index 2025.” Bengaluru: congestion level, time lost per year and travel time per 10 km.

[2] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson and Blaise Agüera y Arcas. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” In: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 54 (2017).



Thank You for Your Attention!

Can traffic be predicted well without sending everyone's raw data to one place?

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